

Supplementary Material

Specification-Aware Fault Diagnosis of ECG Analog Front-Ends: Bridging Alternate Test and Machine-Learning Diagnosis for In-Service Self-Test

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TABLE S1
THE TWO CIRCUITS

	Circuit A	Circuit B
Instrumentation amplifier	INA333	three op-amps
Supply	3.3 V	± 5 V
Gain (first stage / total)	4.03 / 278	10.36 / 414
Resistors / capacitors with faults	16 / 8	20 / 6
Discrete op-amps	5	6
Converter range	0 to 3.3 V	± 2.5 V
Fault conditions	293	307
Simulated cases	63 600	66 400

TABLE S2
SELF-TEST MEASUREMENTS

Set	Measurement	Features	Time
C1	Dc level at the output and at the instrumentation-amplifier output	2	0.13 s
C2	Differential and common-mode tones at 0.05, 0.5, 10, 50 and 150 Hz: gain, phase and CMRR estimate	15	94 s
C3	Response to the 1 mV, 200 ms calibration pulse: baseline, peak, final value, undershoot, tail, rise time, area	7	1 s
C4	Response to a 1 nA lead-off current at 10 and 100 Hz	2	2 s

TABLE S3
DETECTOR TRAINED ON THE PERCENTAGE LABEL OR ON THE FUNCTIONAL LABEL, SCORED AGAINST COMPLIANCE. MEAN OF THREE REPETITIONS, IN PERCENT

Circuit	Trained on	Labeled bad	Escapes	False rejects
A	Percentage	88.7	7.9	36.8
	Functional	21.0	4.1	1.6
B	Percentage	89.2	8.3	36.5
	Functional	30.2	7.6	2.2

TABLE S4
LOCATION OF THE CAUSE ON NON-COMPLIANT CASES: MACRO F1 (MEAN OF THREE REPETITIONS)

Model	By ambiguity group		By component	
	A	B	A	B
<i>k</i> -nearest neighbors	0.83	0.70	0.67	0.55
Random forest	0.86	0.76	0.74	0.61
Support vector machine	0.86	0.71	0.69	0.53
Gradient boosting	0.87	0.74	0.73	0.59
Multilayer perceptron	0.81	0.71	0.66	0.53
Convolutional network ^a	0.56	0.32	–	–

^aOn the sampled pulse response only.

TABLE S5
ORIGIN CLASSIFICATION ON THE TEST PART OF ONE REPETITION (RANDOM FOREST, MAIN FEATURE SET)

Circuit	True	Predicted			Recall
		None	Circuit	Electrode	
A	None	14 314	63	60	99.2 %
	Circuit	189	3649	25	94.5 %
	Electrode	269	15	496	63.6 %
B	None	13 029	136	68	98.5 %
	Circuit	445	5417	45	91.7 %
	Electrode	278	25	477	61.2 %

- I. CIRCUITS AND MEASUREMENTS
- II. FUNCTIONAL AGAINST PERCENTAGE SEVERITY
- III. LOCATION AND ORIGIN
- IV. ROBUSTNESS AND MINIMAL SET OF MEASUREMENTS

This document gives the tables and figures behind the results that the main text reports in summary. Circuit A is the front-end built around an integrated instrumentation amplifier and Circuit B the discrete three-op-amp front-end. Every number comes from the open dataset and code of the article.

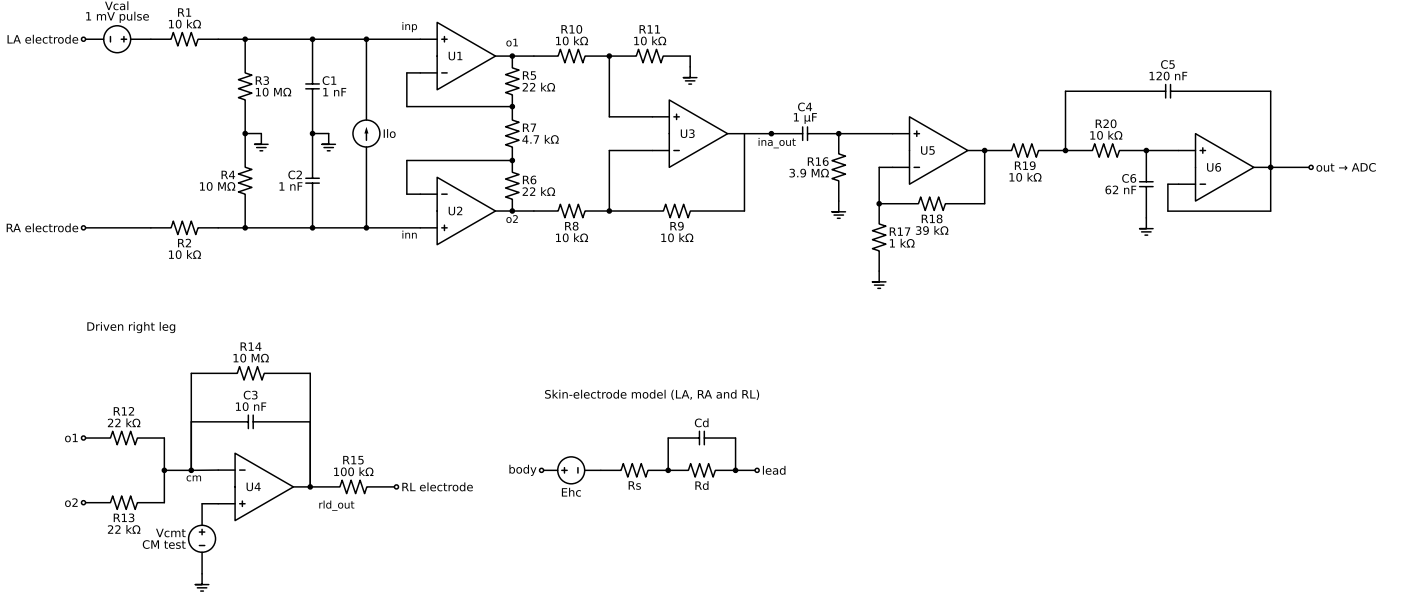


Fig. S1. Circuit B: discrete three-op-amp instrumentation amplifier on ± 5 V supplies, with the same filters, gain stage and driven-right-leg integrator as Circuit A.

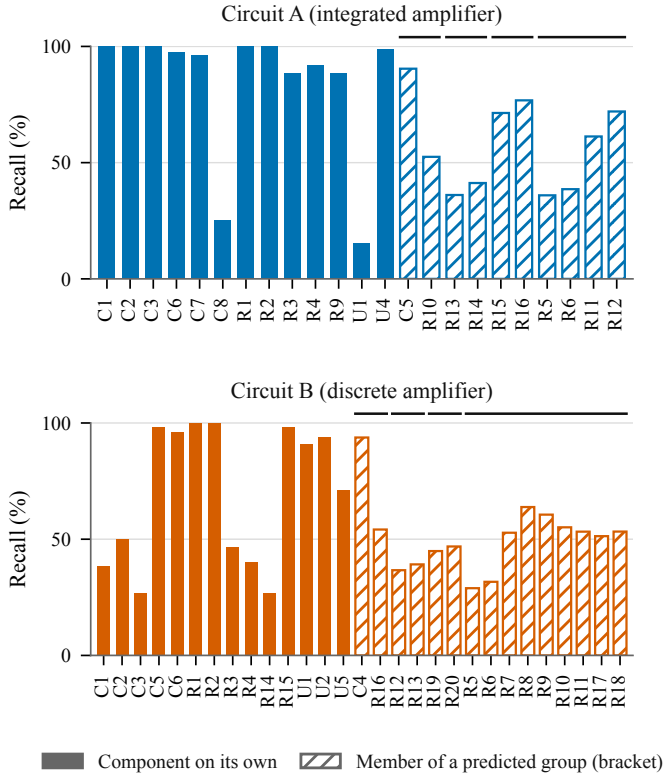


Fig. S2. Recall of each component by the best component-level model (random forest) on non-compliant cases. Hatched bars are members of the ambiguity groups predicted from the sensitivities of the nominal circuit; a bracket joins the members of each group.

TABLE S6
ROBUSTNESS OF THE COMPLIANCE DECISION (CLASS-WEIGHTED CLASSIFIER) AND OF GROUP-LEVEL LOCATION. MEAN OF THREE REPETITIONS

Condition	Circuit A			Circuit B		
	Esc.	F.r.	F1	Esc.	F.r.	F1
Default (1 mV, 12 bits)	4.1	1.7	0.88	7.6	2.2	0.77
10 mV noise, matched training	4.2	1.6	0.86	7.8	2.2	0.75
10 mV noise, trained at 1 mV	9.7	1.2	0.79	6.6	18.8	0.71
8 bits, matched training	3.9	1.6	0.88	7.5	2.2	0.76
8 bits, trained at 12 bits	7.0	1.2	0.87	7.2	4.2	0.75
Truncated Gaussian tolerances	3.8	1.6	0.89	7.4	2.4	0.77
2% resistors, 10% capacitors	7.1	1.9	0.87	11.4	2.9	0.45

Esc.: escapes (%). F.r.: false rejects (%). F1: macro F1 of the location by ambiguity group. The last two rows are scored on separately generated datasets.

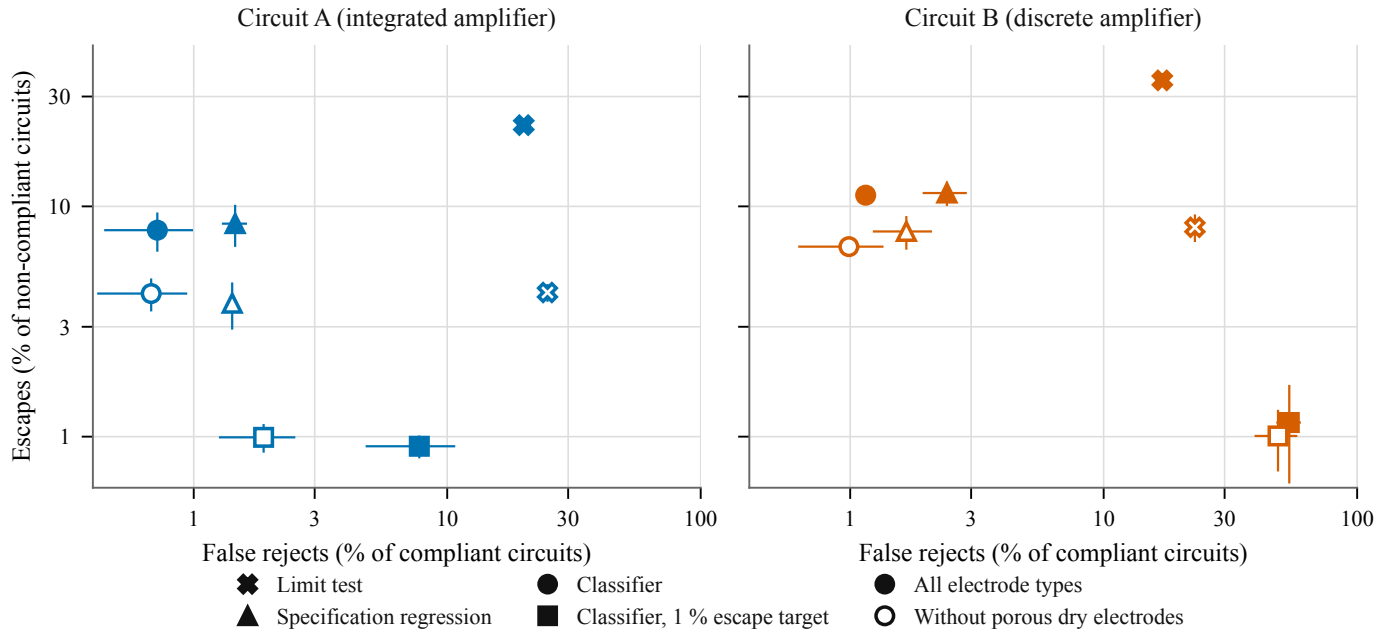


Fig. S3. Operating points of the four decision methods with 95 % confidence intervals over three repetitions, for all electrode types (filled markers) and without porous dry electrodes (open markers).

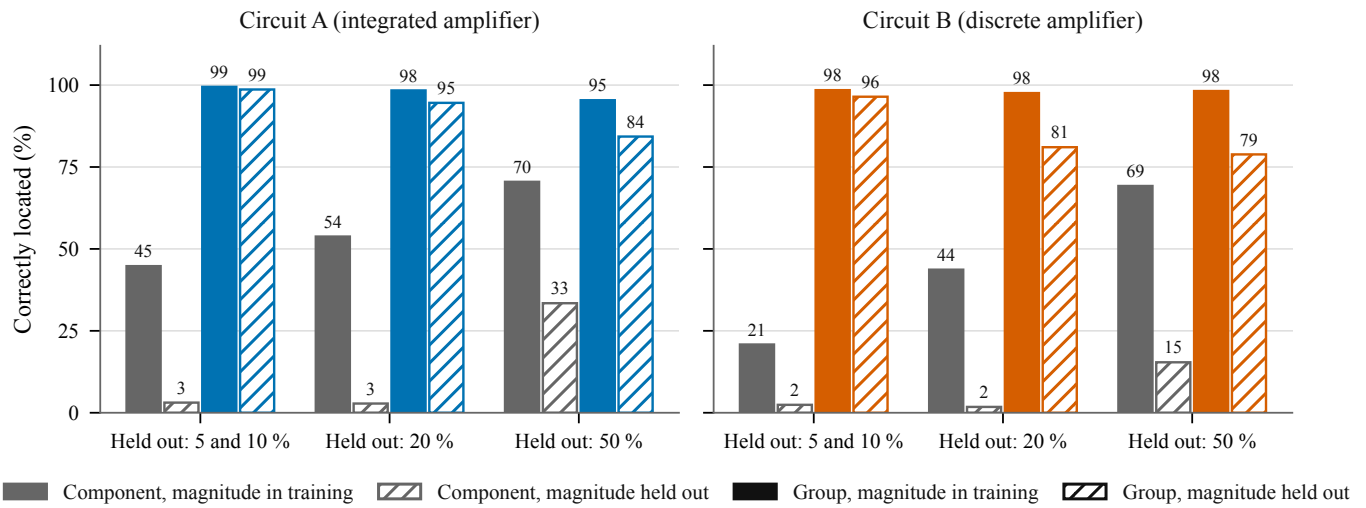


Fig. S4. Location of the cause for parametric faults of magnitudes present in training or held out of it, scored on the non-compliant cases of those magnitudes.

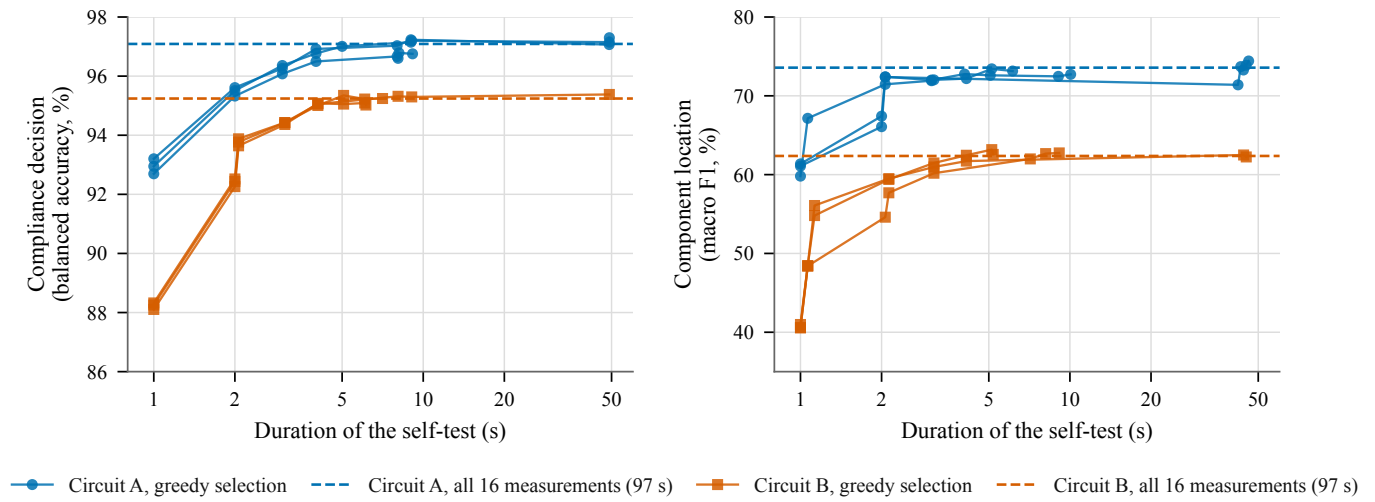


Fig. S5. Score on the test part as self-test measurements are added one at a time by greedy selection, for the compliance decision and for component location. One line per repetition; dashed lines are the score with all the measurements.